

Practical 6

Aim: Displaying Time over 4-Digit 7-Segment Display using Raspberry Pi.

Connection scheme Raspberry Pi

TM1637 Board Pin	Function	RPI Pin	Raspberry Function
GND	Ground	14	GND
VCC	+5V Power	4	5V
DI0	Data In	18	GPIO 24
CLK	Clock	16	GPIO 23

Step 1: Open the terminal and enter the following command.

```
pip install RPi.GPIO --break-system-packages
```

To control the LED, we use a special script with pre-defined functions. Various functions are available in the script, for example you can display numbers and adjust the intensity of the LEDs. Download the script with the command:

<https://github.com/bhoomikapansare/7Segment.git>

Step 2: Type the following python program in Thonny.

```
import sys
```

```
import time
```

```
import datetime
```

```
import RPi.GPIO as GPIO
```

```
import tml1637
```

```
GPIO.setmode(GPIO.BCM) # or GPIO.BOARD depending on your wiring
```

```
Display = tml1637.TM1637()
```

```
Display.Clear()
```

```
Display.SetBrightness(1)
```

```
while True:
```

```
    now = datetime.datetime.now()
```

```
    hour = now.hour
```

```
    minute = now.minute
```

```
    second = now.second
```

```
currenttime = [int(hour / 10), hour % 10, int(minute / 10), minute % 10]
```

```
Display.Show(currenttime)
```

```
Display.ShowDoublepoint(second % 2)
```

```
time.sleep(1)
```